



A Hundred Thousand Welcomes to Ireland, Dublin and Tallaght.

Large portions were generated by an AI in “conservation” with a human. G Stockil 28th June 2026

Contents

Thanks for arriving in Ireland.....	2
DAY1 Talk	3
SCHEDULE ---.....	3
Background.	4
COLLEGE PC.....	5
Microsoft Copilot is installed on all college PCs. It may be useful for generating AI code, although I have not tested it for this purpose. If you already have a subscription to any of the other GAI tools, consider using it.....	5
QUIZ.....	5
Prompting AI Effectively: Give It the Full Picture	5
WIFI ACCESS ON TU DUBLIN	7
What is in the Kit Boxes?	7
Contents of the Kit Box.....	7
⚠️ HARDWARE SAFETY NOTES — Please Read Before Touching the Boards	8
Power Bank Design with AI Prompts.....	8
SOME PHOTOS	12
YOU.....	14
NMEA READER PROGRAM	15
Reading NMEA: Before and After GNSS Lock · Waveshare LC29H(AA) + TenStar ESP32-S3 · SENSATE-X 2.0 / STMP26.....	16
1. Introducing the LC29H(AA): A Chip That Does the Hard Work	17
1.1 What Makes the LC29H(AA) Special?	17
1.2 Why NMEA Makes Complexity Disappear.....	17
1.3 Common NMEA Sentence Types You Will See.....	18
2. The Bench Testing & Program Download Flowchart.....	19
3. Expected Console Output Log	20
4. Laboratory Troubleshooting Checklist.....	22
4.1 The Bootloader Amnesia (Skipping Step 2).....	22
4.2 The Ghost Port Shuffle (Skipping Step 3).....	22
4.3 The Stuck Bootloader (Skipping Step 5).....	22

4.4 The Empty Data Panic (Skipping Step 7).....	22
5. Waveshare GNSS Board Diagnostic LED Reference.....	22
6. Technical Glossary & Concepts	24
7. Suggested Open-Source Graphics & Visualisation Resources	24
7.1 GNSS Satellite Geometry & Constellation Diagrams	24
7.2 NMEA Sentence Structure Diagrams	25
7.3 Hardware Block Diagrams & Signal Flow	25
7.4 General Scientific Diagram Tools	25
APPENDIX B NMEA_Reader Program.....	28



Céad Míle Fáilte go hÉirinn, go Baile Átha Cliath agus go Tamhlacht.



STMP – SENSATE SUMMER SCHOOL
DUBLIN 29TH JUNE – 3RD JULY 2026
 SENSATE Sensors for Applications in Technological Education

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Thanks to EUT+ for funding this summer school and for enabling student activity between these institutes, and other non EUT+ institutes who are special guests.

A Hundred Thousand Welcomes to Ireland, Dublin and Tallaght.

Thanks for arriving in Ireland

DAY1 Talk

SCHEDULE ---

**

STMP26 SENSATE-X 2.0 Programme Schedule

Lunch is daily from **13:00 – 14:00**. (TEL visitors will have lunch provided at TEL)

First Day (Monday 29th June)

09:00 – 09:30 Early Arrivers 09:30 – 10:00 Coffee (Staff Room)
 10:00 Formal Start – Welcome by Head of Department James Wright 10:10 Introductory Talk
 10:15 Talk by TEL (Tokyo Electron) – Joanna Orlinska 10:30 Technical Details and Your Kits
 11:00 – 11:10 Short Break
 11:10 – 12:00 Student-Led Presentations and Activities 12:00 – 13:00 Teams Meet and Plan
 13:00 – 14:00 Lunch 14:00 – 17:00 Project Work (take short breaks as needed)

Tuesday (30th June)

09:00 Bus departs for TEL (Tokyo Electron) – Teams 1 to 5 09:00 – 14:00 TEL Company Visit
 (lunch provided at TEL) 14:00 Return to campus 14:00 – 17:00 All groups – Continue Project
 Work
(Teams not visiting TEL will continue project work as normal)

Wed (1st July ²⁰²⁶)

09:00 Bus departs for TEL (Tokyo Electron) – Teams 6 to 10 09:00 – 14:00 TEL Company Visit
 (lunch provided at TEL) 14:00 Return to campus 14:00 – 17:00 All groups – Continue Project
 Work
(Teams not visiting TEL will continue project work as normal)

Thursday (2nd July ²⁰²⁶)

09:00 Bus departs for TEL (Tokyo Electron) – Teams 11 to 16 09:00 – 14:00 TEL Company
 Visit (lunch provided at TEL) 14:00 Return to campus 14:00 – 17:00 All groups – Continue
 Project Work
(Teams not visiting TEL will continue project work as normal)

Friday

09:00 – 10:00 Finalise Presentations 10:00 Bus departs for Surprise Visit (return around 14:00)
 13:00 – 14:00 Lunch
 14:00 Team Presentations (Max 5 slides and 5 minutes per team)
 15:00 Certificate Ceremony

Background.

Many of you have just finished exams — some very recently, and this was your focus. In June we had a Blended (online) phase where you met your team. Hopefully you had time to do the quizzes, meet a few times and do some reading, if you did not, there is plenty of time to do it now and we are here to work together.

We have around 16 teams one or two smaller teams may be merged if you wish , most teams are 4 or 5 people.

#	Gaeilge	Română	Español	Italiano	Deutsch	English	#
1	A hAon	Unu	Uno	Uno	Eins	One	1
2	A Dó	Doi	Dos	Due	Zwei	Two	2
3	A Trí	Trei	Tres	Tre	Drei	Three	3
4	A Ceathair	Patru	Cuatro	Quattro	Vier	Four	4
5	A Cúig	Cinci	Cinco	Cinque	Fünf	Five	5
6	A Sé	Şase	Seis	Sei	Sechs	Six	6
7	A Seacht	Şapte	Siete	Sette	Sieben	Seven	7
8	A hOcht	Opt	Ocho	Otto	Acht	Eight	8
9	A Naoi	Nouă	Nueve	Nove	Neun	Nine	9
10	A Deich	Zece	Diez	Dieci	Zehn	Ten	10
11	A hAon Déag	Unsprezece	Once	Undici	Elf	Eleven	11
12	A Dó Dhéag	Doisprezece	Doce	Dodici	Zwölf	Twelve	12
13	A Trí Déag	Treisprezece	Trece	Tredici	Dreizehn	Thirteen	13
14	A Ceathair Déag	Paisprezece	Catorce	Quattordici	Vierzehn	Fourteen	14
15	A Cúig Déag	Cincisprezece	Quince	Quindici	Fünfzehn	Fifteen	15
16	A Sé Déag	Şaisprezece	Dieciséis	Sedici	Sechzehn	Sixteen	16

We have **16 boxes**, each containing pre-built kits. There is **one kit per team**, with additional kits available if you would like to carry out extra experiments. Some assembly or soldering may be required for the extra kits.

To help you get started quickly, each team will receive **one pre-built kit**. Every kit has a **green**

heart sticker.  **When opening your kit keep the boards in the electrostatic bag in the kit**

The main laboratories are **Rooms 215, 217, and 219**. There are also several smaller rooms nearby. Teams are welcome to work in the laboratories, the smaller rooms, or the surrounding study areas. We assume each team has access to at least one laptop.

COLLEGE PC

It is best you use your own laptop , however you can use a TU Dublin PC with a temporary shared login. As this password is shared by all participants, **please do not store any sensitive or personal data on the college PCs. Use the following Username and Password.**

Username: TA.Guest@tudublin.ie
Password: Eng_Summer_2026

Note that you are unlikely to be able to use a college PC for all aspects of the TenStar board, as it **NMEA_Reader** program as it does not use specialised libraries. Most other code will use specialised libraries.

Microsoft Copilot is installed on all college PCs. It may be useful for generating AI code, although I have not tested it for this purpose. If you already have a subscription to any of the other GAI tools, consider using it.

QUIZ

When I say TenStar board what am I referring to? Can TenStar be spelt Tenstar, or does it matter too much?.

When I say Waveshare board what am I referring to?

When I say NMEA what am I referring to ?

Prompting AI Effectively: Give It the Full Picture

Please enter this into your AI, modify it any way you wish. It tells the system a lot about the specific hardware you are using and avoids many (but not all) possible errors. Get used to prompting AI to

help you and have technical chats with it. If English is not your first language, ask it to give answers in your native language, with or without an English translation, as you like.

Do not be passive — push back on AI answers, verify its outputs, and treat it as a conversation rather than an oracle. AI can be wrong, and your own judgement and understanding are what matter in the end.

You have the probably second most powerful thought tool ever invented — the most powerful one is you, and never forget that. AI may seem human, **but only you are truly human and have the ability to have genuine insight and 'humanity'.**" The prompt is on a grey background

Act as a supportive, expert pedagogical guide and university professor teaching embedded programming and satellite navigation systems. Your target audience consists of engineering students who possess a foundational knowledge of C++ but require structured guidance, clear analogies, and deep insights when mastering real-time hardware integration. You possess expert-level knowledge of Global Navigation Satellite Systems (GNSS) and an intricate, field-level understanding of NMEA 0183 sentence structures, talker IDs, and data field layouts.

Generate a complete, fully compilable modular Arduino style C++ program for an ESP32-S3 TenStar development board physically interfaced with a multi-constellation Quectel LC29H (AA) GNSS receiver module.

The program must be configured to the following hardware-specific engineering specifications:

1. UART Communication: Initialize Hardware Serial 2 at its native default speed of 115200 baud (8N1 configuration) on Pins RX=17 and TX=18 to accurately handle the high-speed multi-system data streams.
2. Core Parsing Engine: Utilize the TinyGPS++ library framework to naturally decode the modern NMEA 4.10 unified multi-system talker ID prefixes (\$GN strings) generated by the LC29H chip.
3. TFT Screen Layout: Render a responsive, flicker-free dashboard using the Adafruit_GFX and Adafruit_ST7789 libraries. The screen must be locked to wide-screen Landscape Mode (Rotation 3) matching the 240x135 dimensions of the TenStar module.
4. Hardware Mapping Pins: Explicitly use the following pin assignments to avoid pin-mapping compiler panics:
 - TFT_CS = 7, TFT_DC = 39, TFT_RST = 40, TFT_backlight = 45
 - SPI_SCK = 36, SPI_MOSI = 35, SPI_MISO = 37
 - Onboard NeoPixel LED_PIN = 33 (1 Element)
5. Library Dependencies: The libraries utilized in this code must be drawn exclusively from the ones mentioned above (Adafruit_GFX, Adafruit_ST7789, Adafruit_NeoPixel, and TinyGPS++). Ensure no deprecated or alternative display frameworks are introduced during code generation.

Pedagogical, Output, and Debugging Rules:

- Never use code placeholders or truncate lines (e.g., do not write "// Insert code here"). Provide 100% complete, ready-to-compile sketches.
- Provide clear, inline annotations explaining the underlying embedded engineering concepts (such as non-blocking execution vs. blocking delays).
- Maintain an encouraging, instructional tone. If the student provides a compiler error or unreadable serial stream in subsequent turns, analyze it methodically, explain the root cause, and guide them to the correct patch.
- Active Debugging Mentorship: Explicitly instruct and guide the student on how to systematically solve unexpected behavior using a combined human-and-AI diagnostic workflow:
 1. AI Technical Chat: Teach them to copy and paste raw terminal error logs, baud rate anomalies, or freezing symptoms into the AI window to get an immediate logical breakdown of the code mismatch.
 2. Peer Collaboration & Human Verification: Encourage them to cross-reference their physical lab bench behavior with neighboring teams. If an identical program works on Bench A but displays zero

satellites on Bench B, guide them to discuss human factors like window proximity, un-tightened SMA antenna gold rings, or loose hardware jumpers.

3. **Signal Isolation:** Show them how to isolate software problems from hardware issues using physical tools, such as observing the flashing onboard NeoPixel to verify that raw data streams are electrically arriving even if data is not yet drawing on the screen.

WIFI ACCESS ON TU DUBLIN



For WiFi access use EduRoam.

or any solution wish.

For AI it is assumed you are signed up to one of the AI tools — use any one you wish; Gemini, Claude, and ChatGPT all work. To set the context of the AI prompt, please feed the prompt described above.

What is in the Kit Boxes?

I will go through the box contents with you — there should be 16 boxes, one per team. You may build more if you wish.

Contents of the Kit Box

1. **Anti-static bags** – Place all logic boards on these whenever they are not in use. This will improve the reliability of the system and reduce possible damage from ESD.
2. **Pre-assembled receiver module**, consisting of:
 - **Waveshare receiver board** (containing the **Quectel** GNSS module)
 - **TenStar board** (containing the **ESP32-S3**)
 - The three serial interface wires are already connected.
 - The short antenna lead is already connected to the IPX connector. There is no need to change the jumpers.
 - Simply attach the external antenna using the screw connector.
3. **USB Type-C cable** – Use this to power and program the TenStar board. Connect it to a USB port on your laptop.
4. **Micro-USB cable** – Use this to power the Waveshare receiver board. Connect it to a second USB port on your laptop or, if one is not available, to an external USB power supply.

5. **3 Meter L Bander GNSS Aerial.** This connects via a screw-thread SMA connector to an IPX connector on the Waveshare board. The IPX connection is already made.

HARDWARE SAFETY NOTES — Please Read Before Touching the Boards

1. ESD — Electrostatic Discharge. These boards are sensitive to static electricity. Always place the boards on the anti-static (silver/pink) bag from your kit before handling. Touch the bag or an earthed metal surface before picking up any board. Do NOT work directly on a wooden desk, clothing, or carpet. A single static discharge you cannot feel can permanently damage the GNSS module or the ESP32-S3.

2. Do NOT use the anti-static bag as a live work surface. The metallic (silver) anti-static bag is slightly conductive. Never place a powered board directly on top of it — this can short the pins on the underside of the board. Use it only as a resting surface when the board is unpowered.

3. The IPX / U.FL antenna connector is delicate. The small push-fit IPX (also called U.FL or IPEX) connector on the Waveshare board is pre-connected in your kit. Do not disconnect and reconnect it repeatedly — U.FL connectors are only rated for approximately 30 mating cycles. If you must connect or disconnect it, press straight down to seat it and lift straight up to remove; never lever it sideways. The SMA screw connector on the cable can be disconnected as often as you like.

4. Getting your first satellite lock — antenna near a window. GNSS signals are very weak and cannot penetrate walls or Low-E glass. For your first lock, place the antenna on a windowsill with a clear view of the sky, or pass the cable outside through a slightly open window. Time to First Fix (TTFF) can be 1–3 minutes in a good location; longer if the sky view is obstructed. Once a fix is established it is much faster to re-acquire.

5. Watch the PPS LED for satellite lock confirmation. The green PPS LED on the Waveshare board flashes once per second (1 Hz) only when a valid 3D position fix is established. Until it starts flashing, the NMEA sentences will contain empty fields (,,,) — this is normal. Do not panic; just wait. The TXD LED flickering green confirms the module is powered and transmitting data even before lock.

6. Power bank / portable USB testing. For outdoor or portable testing, a standard USB power bank can power the Waveshare board via its Micro-USB cable, while your laptop (or a second power bank) powers the TenStar board via USB-C. This lets you take the setup outside for better sky view without needing a mains socket. Ensure the power bank provides at least 500 mA at 5 V.

Power Bank Design with AI Prompts.



Two USB power outputs on a power bank can be used to power these devices and make portable measurements. You do look a little unusual — someone carrying a strange circuit with a 3 m antenna around your neck — but it works!

A sample portable unit with a power bank is shown below. It has an AI-assisted program that gives PVT information. I am amazed at how well the AI works (provided it is prompted correctly and you watch out for strange behaviour and question the answers, but it is a fantastic time saver).

Sample Prompt (we will come back to this another time) I will skim it now, read it or similar later

I am a student on the STMP26 SENSATE-X 2.0 Short Term Mobility Programme at TU Dublin Tallaght Campus.

STARTING POINT

I have already completed the NMEA_Reader project. This means:

- My hardware is correctly wired and powered
- NMEA sentences are flowing from the LC29H(AA) into the ESP32-S3
- I can see valid NMEA data in the Arduino IDE Serial Monitor
- I know which COM port my board is on
- My Arduino IDE is correctly configured for the TenStar ESP32-S3

I now want to take the next step and display live PVT data
(Position, Velocity, Time) on the built-in TFT screen.

HARDWARE

Microcontroller: TenStar TS-ESP32-S3 development board

GNSS Receiver: Waveshare LC29H(AA) dual-band GNSS module

Powered by its OWN USB cable — not from the ESP32

External patch antenna connected — always required

Display: Built-in TFT on the TenStar board

WIRING (already confirmed working)

Black Dupont wire — LC29H GND → ESP32 GND
Yellow Dupont wire — LC29H TX → ESP32 GPIO 17 (ESP32 receives)
Blue Dupont wire — LC29H RX → ESP32 GPIO 18 (ESP32 transmits)

PIN DEFINITIONS (confirmed working — do not change)

```
#define GPS_RX      17
#define GPS_TX      18
#define TFT_CS      7
#define TFT_DC      39
#define TFT_RST     40
#define TFT_backlight 45
#define SPI_SCK     36
#define SPI_MISO     37
#define SPI_MOSI    35
```

ARDUINO IDE SETTINGS (Tools menu)

Board: ESP32S3 Dev Module
USB CDC On Boot: Enabled
Board package: Espressif ESP32 (not the Arduino version)

LIBRARIES ALREADY INSTALLED

TinyGPSPlus — parses NMEA sentences from the LC29H
Adafruit GFX — core TFT graphics
Adafruit ST7789 — TFT display driver

WHAT I WANT

- Please write a well-commented Arduino sketch that:
- Uses TinyGPSPlus to parse the incoming NMEA data
 - Displays the following on the TFT screen:
 - P — Position: latitude, longitude, altitude
 - V — Velocity: speed in km/h, heading in degrees
 - T — Time: UTC time and date
 - Also shows satellites in view, HDOP, and fix status
 - Is under 200 lines including comments
 - Is structured clearly so I can see how the program is built:
 - libraries → pins → constants → objects → setup() → loop()
 - helper functions
 - Refreshes the display once per second

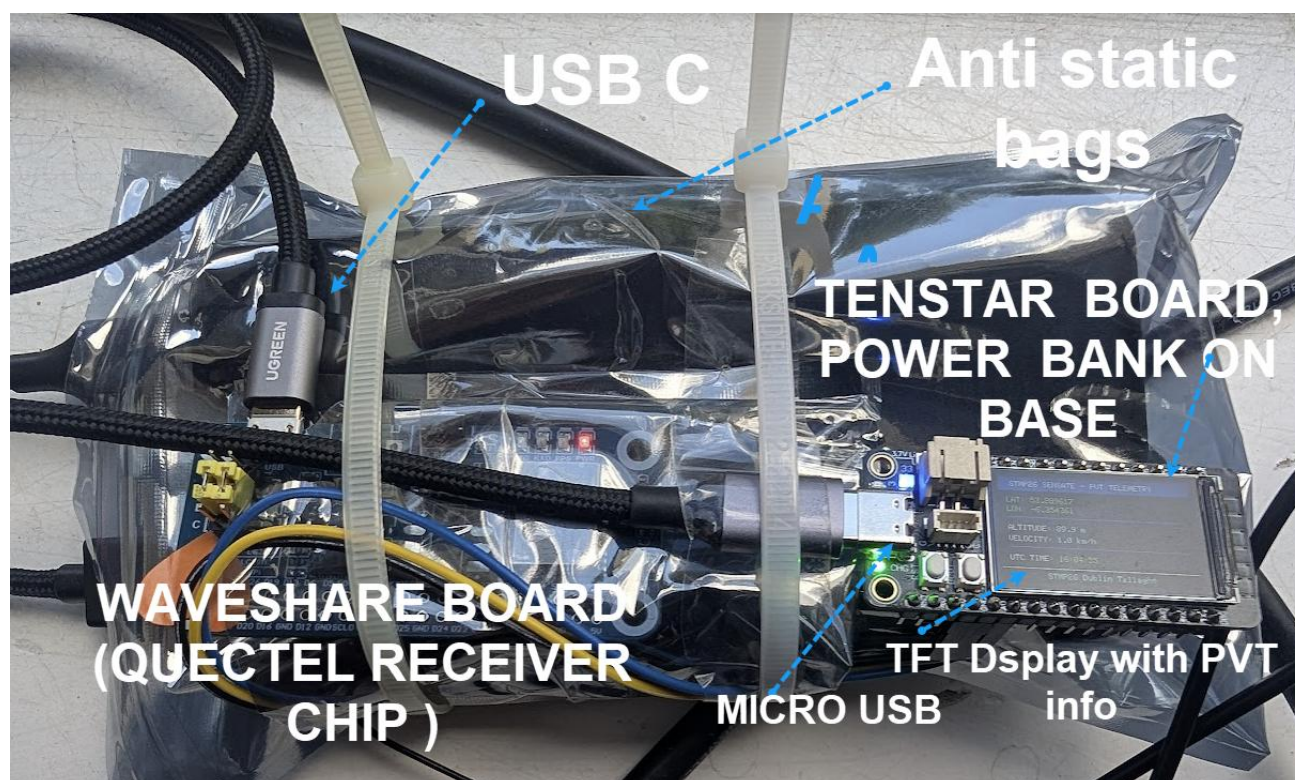
KNOWN TIPS

- Call `digitalWrite(TFT_backlight, HIGH)` BEFORE `tft.init()` or the screen will stay dark
- Use `HardwareSerial GPSSerial(1)` for the GPS UART
- The "Chars processed" count in Serial Monitor confirms data is flowing even before a GPS fix is achieved
- If no Galileo satellites appear, raise the antenna — it needs a clear view of the southern sky from Dublin

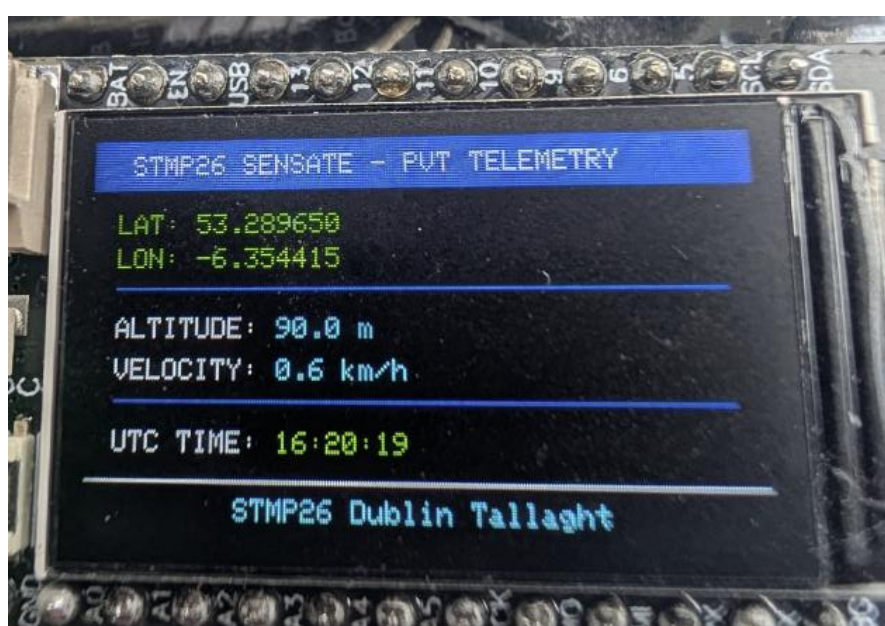
- Press the RST button on the board if the sketch does not start automatically after uploading

Please keep the code simple and well commented. I want to understand how it works, not just copy and paste it.

This is the board connected to the power bank, the emphasis is anti static protection,



PVT Information

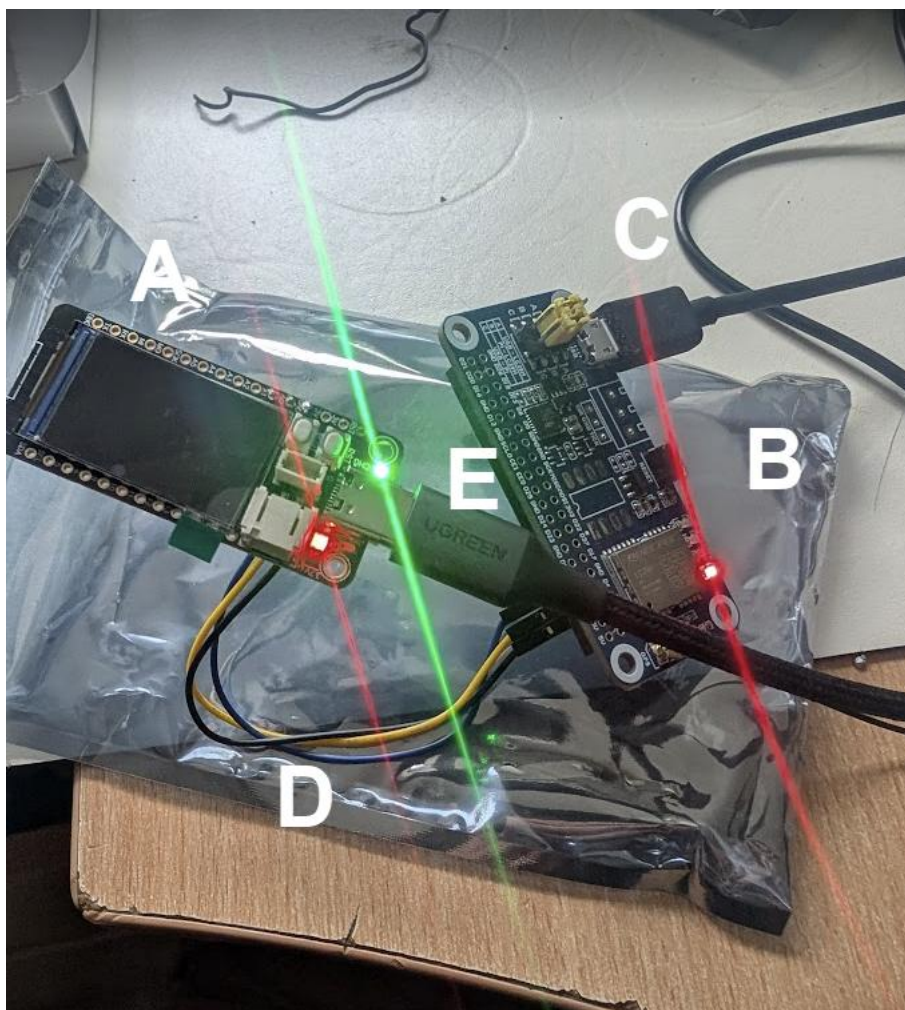


According to Wikipedia , Tallaght is located at the coordinates **53.286° N latitude** and **-6.373° W longitude**, is there a difference?

SOME PHOTOS

In this section I want you, where possible, to look at your kits and answer the following questions — and ask any of your own.

Given the picture shown, provide suitable names for Items A to E.



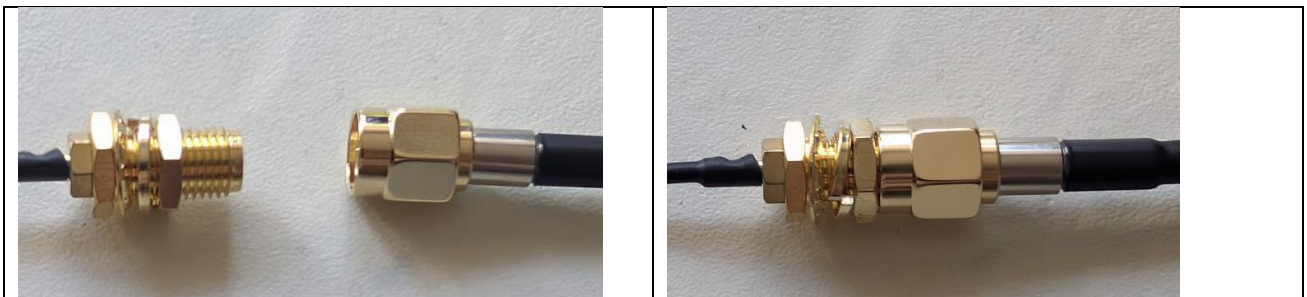
Letter	Suitable Name
A	
B	
C	
D	
E	

Which of these boards has the Quectel LC29H(AA) device, and what might we call this board?

Your IPX connector is already connected



What is the following connector called is connected to the IPX (IPEX, U.FL) connector by a 17 cm cable



As show where does the connector on the left to, and the one on the right. You can disconnect this connector as often as you like, do not change the IPX connection (at least not often)

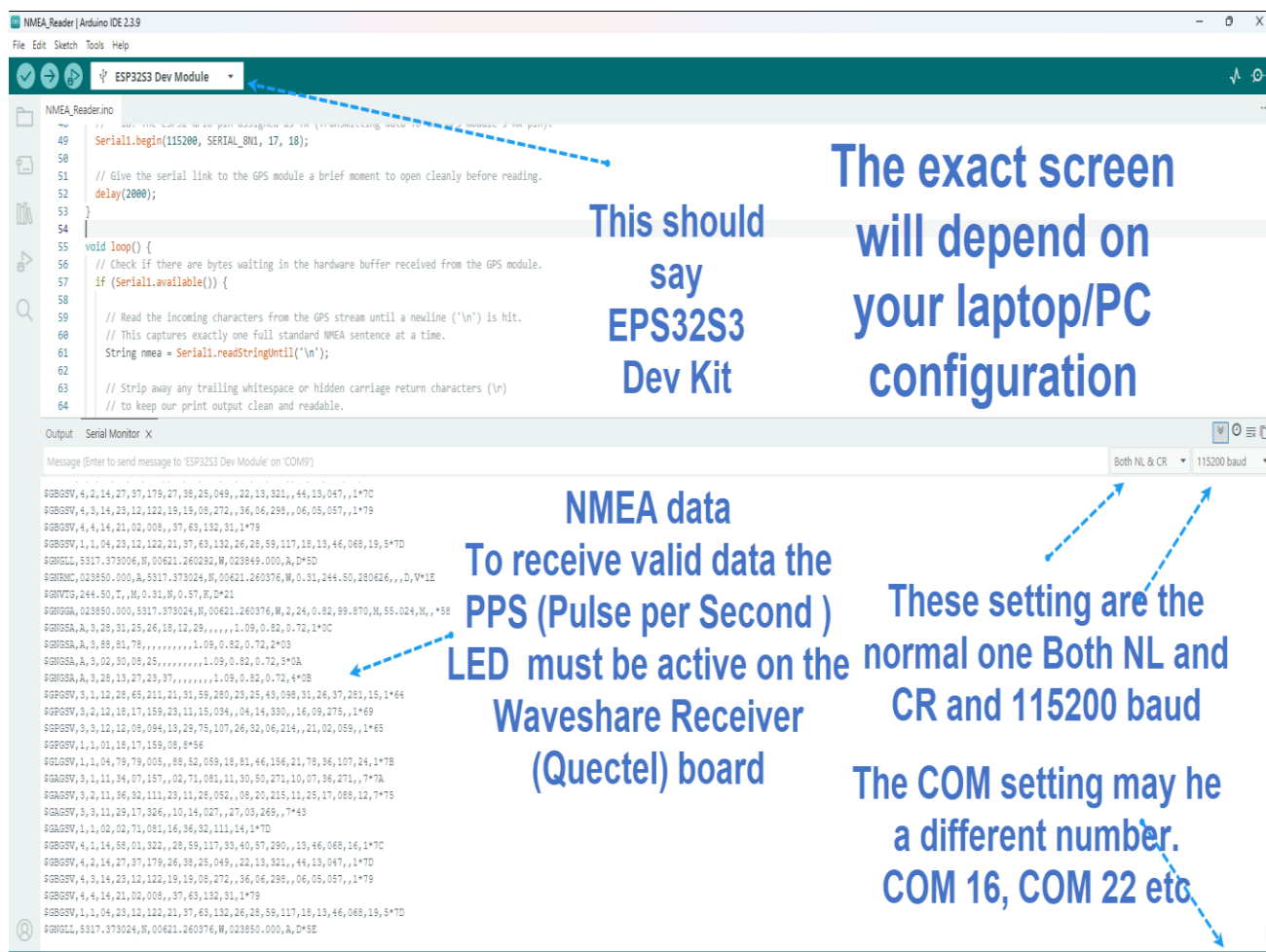
What is this component, sometime called a puck,



YOU

NMEA READER PROGRAM

Find this program at https://github.com/gstockiltud/BIP26Dublin/tree/main/NMEA_Reader



A detailed debug guide is give below here is the general **simplified sequence**

Here's the **simplified** sequence but this often works:

1. Connect TenStar board via USB-C to PC
2. Connect Waveshare board via Micro-USB to PC (or power bank)
3. Hold **BOOT**, press & release **RESET**, release **BOOT**
4. Open the **NMEA_Reader** sketch in Arduino IDE
5. Go to **Tools > Port** and select the active COM port
6. Click the **Upload** arrow and wait for *"Upload Complete"*
7. Press & release **RESET** on the TenStar board
8. Open the **Serial Monitor**
9. Confirm NMEA sentences are scrolling (\$GNGGA, \$GNRMC etc.) and the **TXD LED** is flickering green
10. Place the antenna with a clear view of the sky
11. Wait for the **PPS LED** to flash once per second — this confirms satellite lock
12. ☒ Live coordinates now populate the Serial Monitor

Reading NMEA: Before and After GNSS Lock · Waveshare LC29H(AA) + TenStar ESP32-S3 · SENSATE-X 2.0 / STMP26

⌚ Warming Up — No Fix (fields that change shown in red/amber)	✅ Locked — Valid 3-D Fix (changed fields shown in green)
\$GNGGA,103522.00,,,,,0,03,5.2,,,M,,M,,*45	\$GNGGA,103535.00,5320.4551,N,00615.8230,
\$GNRMC,103522.00,V,,,,,0.00,,N*7C	W,1,09,1.1,84.3,M,-51.4,M,,*52
\$GNGSA,A,1,01,09,,,,,,,,,5.2,5.1,0.8,1*05	\$GNRMC,103535.00,A,5320.4551,N,00615.8230
\$GPGSV,3,1,11,01,25,045,00,09,60,120,14,17,1	,W,0.12,,270626,,N*68
0,220,00,28,35,310,08*73	\$GNGSA,A,3,01,09,17,28,02,12,,,,,1.8,1.1,0.8,1*
\$GAGSV,2,1,05,02,40,070,00,12,55,150,11,24,2	0B
0,260,00*62	\$GPGSV,3,1,11,01,25,045,38,09,60,120,42,17,1
	0,220,35,28,35,310,40*76
	\$GAGSV,2,1,05,02,40,070,37,12,55,150,44,24,2
	0,260,31*65

Key fields that change at lock acquisition

Sentence	Field	No Fix	Fixed	What it means
GGA	Fix Quality (field 6)	0	1	0 = no fix; 1 = GPS/GNSS fix. Lat/Lon fields are empty until this becomes 1.
RMC	Status (field 2)	V	A	V = Void (data unreliable); A = Active. PPS pin only pulses when Status = A.
GSA	Fix Type (field 3)	1	3	1 = no fix; 2 = 2-D fix; 3 = 3-D fix (altitude valid). Altitude in GGA is only reliable at fix type 3.
GSV	SNR (field 7,11...)	00–14	31–44	Signal-to-Noise Ratio in dBHz. Values < 20 are too weak to use; > 30 signals contribute to a fix.

⚡ **PPS Note:** The Pulse-Per-Second (PPS) pin on the LC29H(AA) remains LOW (silent) while RMC Status = **V**. Once a valid 3-D fix is acquired and RMC Status transitions to **A**, the PPS pin starts emitting a 100 ms pulse every second, accurately aligned to UTC. This pulse is the heartbeat that timing-critical applications (e.g. OSNMA authentication, NTP servers) depend on.

SENSATE-X 2.0 — STMP26 Lab Reference · TU Dublin Tallaght · June 2026 · Gerard Stockil

The next section is details, read yourself. Some of it has been covered in previous Training document. Ask your teammates , ask the lecturers and ask AI and the web to help you

STMP26 SENSATE-X 2.0

Student Lab & Troubleshooting Guide

GNSS Bench Testing, Program Download & Satellite Tracking

TU Dublin Tallaght Campus | 29 June – 4 July 2026

1. Introducing the LC29H(AA): A Chip That Does the Hard Work

Before you touch a cable or open the Arduino IDE, it is worth taking a moment to understand the most important piece of silicon on your bench. The Waveshare module at the heart of this lab is built around the Quectel LC29H(AA) — a compact, highly sophisticated GNSS receiver that simultaneously tracks signals from multiple global satellite constellations: GPS (USA), GLONASS (Russia), Galileo (Europe), BeiDou (China), and QZSS (Japan). Despite that complexity under the hood, as a user you will never need to decode raw radio signals or interpret orbital mathematics. The chip handles all of that internally, and communicates its results to you in a single, elegant, human-readable format called NMEA 0183.

1.1 What Makes the LC29H(AA) Special?

The 'AA' suffix in the part number indicates the dual-frequency variant. Consumer-grade receivers typically listen on one radio frequency (L1), but the LC29H(AA) simultaneously receives on both L1 and L5. Receiving two frequencies allows the chip to mathematically cancel out a major source of error — signal delay introduced by the ionosphere — resulting in position accuracy that can reach the sub-metre level outdoors under good sky conditions. Key characteristics at a glance:

- Multi-constellation: GPS, GLONASS, Galileo, BeiDou, QZSS — all active simultaneously.
- Dual-frequency: L1 + L5 reception for superior ionospheric error correction.
- Galileo OSNMA support: cryptographic navigation message authentication — a world first in open-service GNSS security.
- Cold start TTFF (Time to First Fix): typically under 30 seconds with a clear sky view. (This can vary a lot and may sometimes be longer)
- Update rate: up to 10 Hz (10 position fixes per second) — well above the 1 Hz default used in this lab.
- Serial output: standard NMEA 0183 sentences at 115,200 baud — plug-and-play with any microcontroller UART.

1.2 Why NMEA Makes Complexity Disappear

Receiving and decoding satellite signals involves relativistic time corrections, Doppler shift compensation, atmospheric modelling, and pseudo-range calculations — a formidable chain of signal processing. The LC29H(AA) performs all of this invisibly. What it delivers to the TenStar ESP32-S3 over a simple two-wire serial connection (TX and RX) is a stream of plain ASCII text lines, each beginning with a dollar sign and a five-letter sentence identifier. This is the NMEA 0183

standard, defined by the National Marine Electronics Association, and it is the same format used by ship navigators, aircraft avionics, and car sat-navs worldwide.

Example:

```
$GNGGA,123519.00,5319.9876,N,00617.1234,W,1,08,0.9,42.3,M,52.0,M,,*47
```

That single line encodes: the UTC time, latitude, longitude, fix quality, number of satellites in use, horizontal dilution of precision (HDOP), altitude above sea level, and a checksum — all separated by commas, all readable without any decoding software. Your Arduino sketch simply reads each line as a string and splits it at the commas. The sophistication of a €1,000 survey-grade signal processor is reduced, from the software perspective, to a CSV file.

1.3 Common NMEA Sentence Types You Will See

Sentence	Source	Key Data
-----	-----	-----
\$GNGGA	All cons.	Time, Lat/Lon, Fix quality, Altitude
\$GNRMC	All cons.	Time, Status, Lat/Lon, Speed, Date
\$GPGSV	GPS	Satellites in view, elevation, azimuth, SNR
\$GAGSV	Galileo	Galileo satellites in view
\$GLGSV	GLONASS	GLONASS satellites in view
\$GNGSA	All cons.	Active satellites, PDOP, HDOP, VDOP
\$GNVTG	All cons.	Course over ground, speed

The two-letter prefix after the dollar sign identifies the constellation source: GP = GPS, GA = Galileo, GL = GLONASS, GB = BeiDou, GN = combined / multi-constellation.

Note on Jumper B: The Waveshare module ships with a configurable UART interface. For this lab, jumper Position B must be set, which routes the LC29H(AA) TX/RX pins to the header used by the TenStar board (GPIO 17 / GPIO 18 at 115,200 baud). If your Serial Monitor shows no output at all and the TXD LED is dark, check the jumper first.

You should **not** need to change the jumper on the Quectel board. Leave it in its default position unless there are exceptional circumstances.

The receiver (which we simply call the **Waveshare board**) comes with a short antenna lead already fitted. Simply screw the antenna onto the connector.

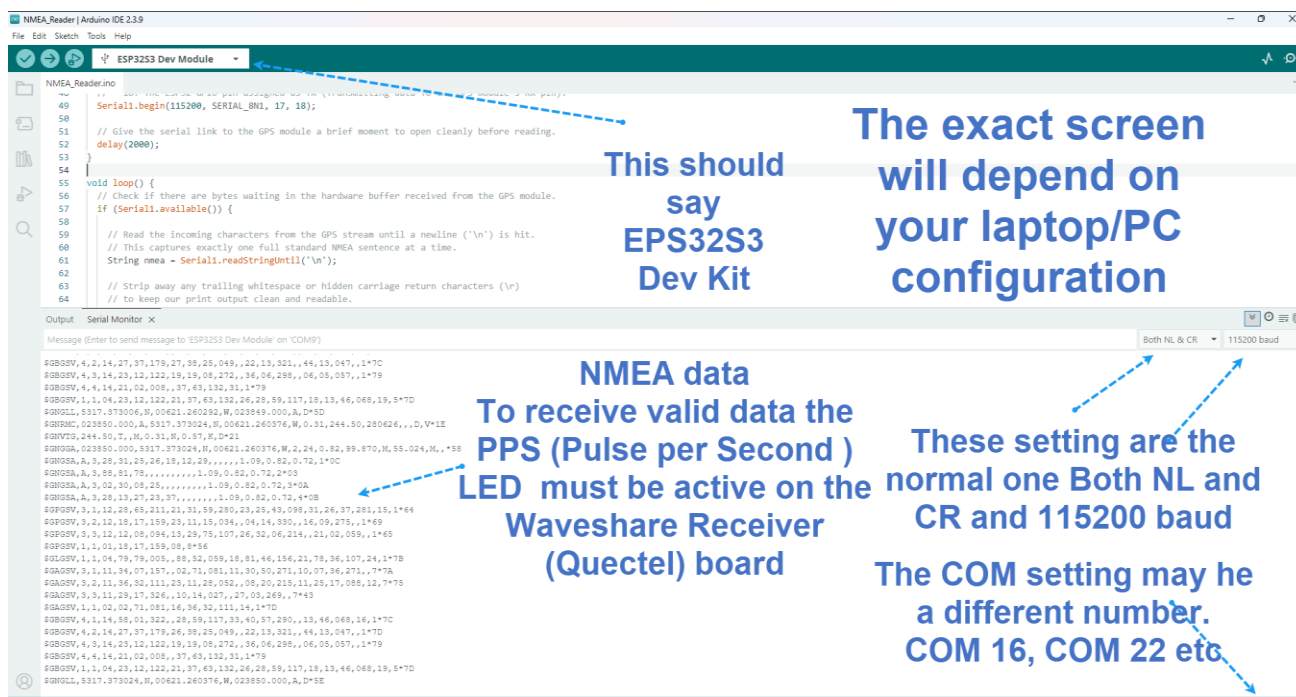
The interface between the **Quectel** module and the **TenStar** board uses three wires. On the pre-built boards, these are already connected, so no further action is required.

2. The Bench Testing & Program Download Flowchart

Use this exact sequence for every new program you run — to verify hardware integrity, flash the program, and confirm serial communications.

```
=====
BENCH TESTING & PROGRAM DOWNLOAD FLOW
=====

[ STEP 1: CONNECT HARDWARE ]
|
+--> TenStar Board    --> USB-C Cable        --> Native PC USB Port
|
+--> Waveshare Board --> Micro-USB Cable    --> Native PC USB Port (Preferred)
                                         OR --> External Power Bank / Wall Plug
|
v  CRITICAL TRAP: Did you skip the button sequence? (See Troubleshooting #1)
[ STEP 2: ENTER PROGRAMMING STATE ]
|
+--> 1. Press & HOLD the physical BOOT button on TenStar board.
+--> 2. Press & RELEASE the physical RESET button.
+--> 3. RELEASE the BOOT button.
|
v  CRITICAL TRAP: Is the wrong port active? (See Troubleshooting #2)
[ STEP 3: INITIALIZE ARDUINO IDE ]
|
+--> Open NMEA Reader program sketch.
+--> Go to Tools > Port and select the active port (e.g. COM22 or COM9).
|
|   +--> CHECK VISUALS: Is TFT Screen completely BLANK?
|               (Normal: bootloader state overrides factory demo.)
v
[ STEP 4: DOWNLOAD PROGRAM ]           (See Expected Console Output, Section 3)
|
+--> Click the Upload arrow icon.
+--> Wait for the console to display: Upload Complete
|
v  CRITICAL TRAP: Did you forget to wake the board? (See Troubleshooting #3)
[ STEP 5: RUN THE PROGRAM ]
|
+--> Press & RELEASE the physical RESET button on the TenStar board once.
|
v
[ STEP 6: DIAGNOSTICS & VERIFICATION ]
|
+--> Open the Serial Monitor.
|
+--> IMMEDIATE CHECK --> Are NMEA sentences scrolling rapidly ($GNGGA etc.)?
|                           AND is the Waveshare TXD LED flickering Green?
|                           (Note: fields will be empty commas initially.)
|
v  CRITICAL TRAP: Is the data empty? (See Troubleshooting #4)
[ STEP 7: SATELLITE FIX ]
|
+--> Ensure antenna has clear sky view (avoid Low-E Glass windows).
+--> WAIT for Waveshare green PPS LED to flash sharply -- once per second.
|
v
[ SUCCESS: Live coordinates and satellite tracking data populate the Monitor! ]
=====
```



3. Expected Console Output Log

When you click Upload in Step 4, the Arduino IDE compiler translates your sketch and streams it to the microcontroller. The output panel at the bottom of the screen should display a sequence that looks exactly like this:

```
Writing '.../tools/partitions/boot_app0.bin' at 0x0000e000...
Flash will be erased from 0x0000e000 to 0x0000ffff...
Compressed 8192 bytes to 47...
Writing at 0x0000e000 [ ] 0.0% 0/47 bytes...
Writing at 0x00010000 [=====] 100.0% 47/47 bytes...
Wrote 8192 bytes (47 compressed) at 0x0000e000 in 0.1 seconds (1186.9 kbit/s).
Verifying written data...
Hash of data verified.

Writing '.../NMEA_Reader.ino.bin' at 0x00010000...
No changed sectors found, verifying if data is in flash...
Note: Data not found in flash, flashing the whole image...
Flash will be erased from 0x00010000 to 0x0005ffff...
Compressed 324752 bytes to 178233...
Writing at 0x00010000 [ ] 0.0% 0/178233 bytes...
Writing at 0x0001c0a4 [=> ] 9.2% 16384/178233 bytes...
Writing at 0x00029522 [====> ] 18.4% 32768/178233 bytes...
Writing at 0x0005f490 [=====] 100.0% 178233/178233
bytes...
Wrote 324752 bytes (178233 compressed) at 0x00010000 in 1.6 seconds (1646.9
kbit/s).
Verifying written data...
Hash of data verified.
Hard resetting via RTS pin...
```

Note: The final line 'Hard resetting via RTS pin...' confirms the computer has finished transmitting code, but the chip remains in suspension until you complete Step 5 manually.

4. Laboratory Troubleshooting Checklist

If something is not working on your bench, 99% of the time the cause is one of these four omissions:

4.1 The Bootloader Amnesia (Skipping Step 2)

- The Arduino IDE upload progress bar hangs indefinitely, or throws the error 'Failed to connect to ESP32-S3: No serial data received'. Symptom:
- You must manually force the microcontroller into programming mode. Hold BOOT, click RESET, then release BOOT. The TFT screen must go blank before you click Upload. Fix:

4.2 The Ghost Port Shuffle (Skipping Step 3)

- The IDE reports 'Serial port not found' or fails to communicate with the board. Symptom:
- Every time the ESP32-S3 enters its native bootloader state, Windows treats it as a brand-new device and reassigns the COM port number. During heavy lab testing these numbers can climb all the way to COM32, before the operating system recycles its counter and drops back to a single-digit port such as COM9. Always verify Tools > Port before every single download. Fix:

4.3 The Stuck Bootloader (Skipping Step 5)

- The IDE console displays 'Upload Complete' or 'Hard resetting via RTS pin...', but the TFT screen stays blank and the Serial Monitor is empty. Symptom:
- The microcontroller is waiting in its programming holding pattern. Press and release the physical RESET button on the TenStar board once. This wakes the chip and starts executing your new code. Fix:

4.4 The Empty Data Panic (Skipping Step 7)

- The Serial Monitor is running but the NMEA strings streaming by show only empty comma-separated fields (e.g. ,,,,). Symptom:
- Do not panic — your hardware is fully functional. The LC29H(AA) outputs blank NMEA rows immediately on power-up to confirm its internal UART engine is active. You must wait for the green PPS LED to establish a satellite lock and begin flashing sharply once per second before coordinates and tracking parameters fill in. Fix:

5. Waveshare GNSS Board Diagnostic LED Reference

LED Label	Colour	State / Behaviour	What It Signifies
PWR	Red	Solid ON	Power Status: The board is receiving steady voltage (VCC).

TXD	Green	Blinking / Flickering Constantly	Data Transmit: The GNSS module is broadcasting raw NMEA sentences immediately on power-up, even without a satellite lock.
RXT	Green	Mostly OFF (Occasional flash)	Data Receive: Blinks only when the TenStar sends a configuration command into the GNSS module.
PPS	Green	OFF initially, then blinks once per second	Pulse Per Second (Lock Status): Flashes sharply with microsecond precision once every second only after a valid 3D position fix is established.

6. Technical Glossary & Concepts

Term	Definition
Almanac	A coarse directory map broadcast by satellites showing the general paths of all active nodes across the network. Valid for months. (From Medieval Latin almanachus / Arabic al-manakh, historically denoting celestial or weather-tracking tables.)
Ephemeris	High-resolution, ultra-precise orbital data unique to each individual satellite, accurate to the metre. Dynamic and expires every 2 to 4 hours. (From Greek ephemeris: 'daily' or 'lasting only a day'.)
Fix	The milestone calculation where a GNSS engine successfully processes signals from at least 4 concurrent satellites to pinpoint its 3D coordinates.
Low-E Glass Shielding	High-efficiency modern window glass embedded with microscopic metal-oxide matrices that unintentionally block RF signals, preventing indoor satellite signal acquisition.
NMEA 0183	The industry-standard serial protocol mapping position, speed, and time parameters into legible ASCII text structures.
TTFF (Time to First Fix)	The duration measured from the moment power is applied to the hardware until a complete geographical position is calculated.
OSNMA	Open Service Navigation Message Authentication. A Galileo-specific security layer that cryptographically verifies the authenticity of navigation signals, protecting against spoofing attacks.
Quectel LC29H(AA)	The GNSS receiver chip on the Waveshare module used in this lab. A high-performance, multi-constellation, dual-frequency receiver supporting GPS, GLONASS, Galileo, BeiDou, and QZSS.

7. Suggested Open-Source Graphics & Visualisation Resources

The following freely available tools and image libraries can be used to generate professional GNSS-related diagrams and illustrations for reports, posters, and presentations. All resources listed below are open-source or carry permissive licences compatible with academic use.

7.1 GNSS Satellite Geometry & Constellation Diagrams

- **The European Space Agency's online Galileo constellation viewer** (<https://gssc.esa.int/navipedia>) includes **SVG-exportable sky plots and constellation geometry diagrams released under Creative Commons. Ideal for showing satellite elevation, azimuth, and coverage maps.**GNSS View (ESA):
- **NASA's open-source Eyes on the Solar System app** (<https://eyes.nasa.gov>) provides **3D animated views of GNSS constellations. Screenshots are freely usable for educational purposes under NASA's media usage policy.**NASA Worldview / Eyes on the Earth:
- **Available on GitHub** (<https://github.com/TheGNSSTools/GNSSCompare>), this project includes **reusable SVG sky-plot components released under Apache 2.0.**GNSS Compare (open-source Android):

7.2 NMEA Sentence Structure Diagrams

- The open-source GPS daemon project maintains a comprehensive NMEA 0183 reference with annotated sentence diagrams. Diagrams are available in SVG format under the BSD licence. GPSTalk project (<https://gpsd.gitlab.io/gpsd/>):
- Search 'NMEA 0183' on <https://commons.wikimedia.org> for CC-BY-SA labelled sentence-structure diagrams suitable for direct inclusion in student documents. Wikimedia Commons:
- u-blox publishes receiver interface descriptions (<https://www.u-blox.com/en/docs>) with clear NMEA field-layout figures. While u-blox-specific, the NMEA figures are generic and widely re-used under fair-use citation in academic work. u-blox Open Documentation:

7.3 Hardware Block Diagrams & Signal Flow

- Waveshare publishes open schematics and block diagrams for all their modules, including the LC29H(AA) breakout board, released for unrestricted educational use. Waveshare Wiki (<https://www.waveshare.com/wiki/>):
- Quectel (<https://www.quectel.com>) provides public-facing hardware design guides and antenna integration drawings under a documentation licence that permits reproduction in academic materials with attribution. Quectel LC29H(AA) Product Page:
- For custom block diagrams, the KiCad EDA project (<https://kicad.org>) provides a massive open-source library of electronic component symbols including ESP32, GNSS modules, and antenna symbols, all under CC-BY-SA 4.0. KiCad Open Symbol Libraries:
- A beginner-friendly open-source PCB and breadboard diagram tool. The Fritzing parts library includes ESP32-S3 and generic UART module components. Export as SVG or PNG for inclusion in Word or PowerPoint. Fritzing (<https://fritzing.org>):

7.4 General Scientific Diagram Tools

- The leading open-source SVG vector editor. Use it to create or modify any of the above diagrams. Fully compatible with LibreOffice Draw and Microsoft Word image import. Inkscape (<https://inkscape.org>):
- A free, browser-based diagram tool with a GNSS and electronics shape library. Export diagrams as SVG, PNG, or PDF. Widely used in academic and industry documentation. draw.io / diagrams.net (<https://app.diagrams.net>):
- For data-driven visualisations such as SNR bar charts or satellite elevation plots from your own NMEA log files. Open-source (PSF licence). See the GNSS-Python community on GitHub for ready-made NMEA plotting scripts. Matplotlib (Python):

Licence reminder: Always check the specific licence of any image before including it in a publicly distributed document. CC-BY and CC-BY-SA require attribution; CC-BY-NC restricts commercial use. When in doubt, cite the source and link to the original.

APPENDIX A — AI Prompt 6th April 2026

Claude Master Prompt Ver A 5th April 2026. Use to contextulise your AI Expenche, can be applied to other GAI. Page 1 of 6 BIP26_CLAUDE_MASTER_PROMPT.txt

===== BIP26 SENSATE-X 2.0 — Master Claude Prompt TU Dublin Tallaght Campus | June 2026 ===== INSTRUCTIONS FOR STUDENTS ----- Copy everything below this line and paste it as your FIRST message to Claude at the start of every programming session. This gives Claude all the hardware and software context it needs so you do not have to re-explain it each time. After pasting this prompt, describe what you want to do — for example: "Write a sketch that displays temperature from the BMP280 on the TFT screen" or "Explain how TinyGPSPlus parses NMEA data". You can refine Claude's output by asking follow-up questions in plain English. Claude will remember the context within a session but not between sessions, so always start with this prompt.

===== PASTE FROM HERE

===== I am a student on the BIP26 SENSATE-X 2.0 Blended Intensive Programme at TU Dublin Tallaght Campus. I am writing Arduino sketches for the following hardware configuration. Please use this information for all code you generate in this session.

----- HARDWARE

----- MICROCONTROLLER

BOARD Name: Tenstar TS-ESP32-S3 Chip: ESP32-S3 FH4R2 (dual core, 240 MHz) Flash: 4 MB PSRAM: 2 MB (disabled in Arduino IDE for BIP26) Product:

<https://www.aliexpress.com/item/1005006590631744.html> Manual:

<https://manuals.plus/ae/1005006590631744> Onboard peripherals: - 1.14 inch ST7789 TFT display (135 x 240 pixels) - BMP280 barometric pressure and temperature sensor (I2C, address 0x77)

Claude Master Prompt Ver A 5th April 2026. Use to contextulise your AI Expenche, can be applied to other GAI. Page 2 of 6 - QMI8658C 6-axis IMU — accelerometer and gyroscope (I2C, address 0x6B) - WS2812 RGB NeoPixel LED (pin 33) NOTE: NeoPixel (pin 33) conflicts with QMI8658C INT pin. Do not use both simultaneously. GNSS MODULE Name: Waveshare LC29H(AA) GPS/DR/RTK HAT Module: Quectel LC29H(AA) dual-band GNSS Wiki:

[https://www.waveshare.com/wiki/LC29H\(XX\)_GPS/RTK_HAT](https://www.waveshare.com/wiki/LC29H(XX)_GPS/RTK_HAT) Baud rate: 115200 (default, 8N1)

Output: NMEA 0183 sentences at 1 Hz default Constellations supported: GPS, Galileo, GLONASS, BeiDou, QZSS ANTENNA — ALWAYS REQUIRED: An external GNSS antenna MUST be connected to the IPX/U.FL connector on the LC29H board at all times. Without it there will be no satellite signal. The connector is a push-fit U.FL / IPEX type — press straight down until it clicks. Place the antenna near a window or outdoors for best results. POWER — IMPORTANT: The LC29H board is powered by its OWN USB Type-C cable. Do NOT power it from the Tenstar 3.3V pin. Both boards connect to the same USB hub on the same PC. JUMPER SETTING: The yellow jumper on the LC29H board must be set to Row B (the PI position). This routes the LC29H UART through the 40-pin Raspberry Pi style header so the Tenstar can read it. If the jumper is on Row A (factory default) nothing will work.

----- WIRING — TENSTAR TO

LC29H (THREE WIRES ONLY)

----- Wire LC29H Pin Tenstar

Front Label GPIO Direction ----- Black Pin 6 (GND) GND GND Common ground Yellow Pin 8 (Pi TXD0) A1 GPIO 17 LC29H transmits → Tenstar RX Blue Pin 10 (Pi RXD0) A0 GPIO 18 Tenstar TX → LC29H receives PIN NAMING

NOTE: The Tenstar board labels pins differently on each side: Front label A1 = Back label 17 = GPIO 17 (used as UART RX) Front label A0 = Back label 18 = GPIO 18 (used as UART TX) These

are analogue-capable pins configured as UART in our sketches. Always use GPIO numbers (17, 18) in Arduino code. Claude Master Prompt Ver A 5 th April 2026. Use to contextulise your AI Expeance, can be applied to other GAI. Page 3 of 6

PIN DEFINITIONS — USE

THESE IN ALL SKETCHES

Source: Mike Gill (TU Dublin) setup guide, confirmed working. // TFT Display #define TFT_CS 7 #define TFT_DC 39 #define TFT_RST 40 #define TFT_backlight 45 // digitalWrite HIGH to turn on // SPI Bus (custom pins — not default ESP32-S3 SPI pins) #define SPI_SCK 36 #define SPI_MISO 37 #define SPI_MOSI 35 // I2C Bus (BMP280 and QMI8658C) #define I2C_SDA 42 #define I2C_SCL 41 // RGB NeoPixel LED #define LED_PIN 33 // Conflicts with QMI8658C — do not use both // GPS UART #define GPS_RX 17 // Receives NMEA from LC29H TX (Yellow wire) #define GPS_TX 18 // Sends to LC29H RX (Blue wire) #define GPS_BAUD 115200 // Sensor I2C addresses #define BMP_ADDR 0x77 #define QMI_ADDR 0x6B

TFT DISPLAY

Driver: ST7789 Size: 135 x

240 pixels (portrait) Landscape: tft.setRotation(3) gives 240 wide x 135 tall Library: Adafruit_ST7789 (NOT TFT_eSPI) Object: Adafruit_ST7789 tft = Adafruit_ST7789(TFT_CS, TFT_DC, TFT_RST); CRITICAL INITIALISATION SEQUENCE (must follow this exact order): SPI.begin(SPI_SCK, SPI_MISO, SPI_MOSI, TFT_CS); // BEFORE tft.init() Wire.begin(I2C_SDA, I2C_SCL); Claude Master Prompt Ver A 5 th April 2026. Use to contextulise your AI Expeance, can be applied to other GAI. Page 4 of 6 pinMode(TFT_backlight, OUTPUT); digitalWrite(TFT_backlight, HIGH); tft.init(135, 240); tft.setRotation(3); tft.fillScreen(ST77XX_BLACK); Initialise the NeoPixel AFTER the display — doing it before can interfere with the SPI bus initialisation.

LIBRARIES — ALL MUST

BE INSTALLED

Library	How installed	Purpose
Adafruit ST7735 and ST7789	Direct install	TFT display driver
Adafruit GFX	Library Auto (dependency)	Graphics drawing primitives
Adafruit BusIO	Library Auto (dependency)	I2C and SPI bus support
Adafruit NeoPixel	Direct install	RGB LED
Adafruit BMP280	Library Direct install	Pressure/temperature sensor
Adafruit Unified Sensor	Library Auto (dependency)	Sensor abstraction layer
TinyGPSPlus	Direct install	NMEA parser for GPS data
SensorLib	Library Direct install	QMI8658C IMU support
SD	Library Auto (dependency)	SD

Installed via SensorLib When installing, always click INSTALL ALL to get dependencies.

ARDUINO IDE SETTINGS

(Tools menu) Board: ESP32S3 Dev Module Flash Size: 4MB (32Mb) ← critical, must match hardware Upload Mode: UART0 / Hardware CDC USB Mode: Hardware CDC and JTAG USB CDC On Boot: Enabled PSRAM: Disabled Upload Speed: 921600 Board package: esp32 by Espressif Systems (NOT the Arduino version — always choose Espressif)

KNOWN BEHAVIOURS

AND QUIRKS

RST

BUTTON: Claude Master Prompt Ver A 5 th April 2026. Use to contextulise your AI Expeance, can be applied to other GAI. Page 5 of 6 After uploading a sketch press the RST button on the Tenstar board to start it running. On some uploads it starts automatically but not always. Always press RST if nothing happens after upload completes. COM PORT CHANGES: The COM port number changes between sessions and sometimes between upload and Serial Monitor. Go to Tools → Port and select the port that appears after plugging in the board. Unplugging and replugging the USB cable and checking again usually resolves any port not found errors. GALILEO SATELLITE COUNT: Galileo may show zero satellites when indoors near a window. This is a sky geometry issue — move the antenna closer to the window and wait up to 2 minutes. Galileo has fewer satellites than GPS so it is more sensitive to partial sky view. TFT STATIC DAMAGE: The TFT display is sensitive to electrostatic discharge. Always work on the anti-static mat and touch it before handling the board. A completely dark screen despite the sketch running normally (NeoPixel working, Serial Monitor printing) is the typical symptom of ESD damage.

STANDARD INCLUDES

FOR A TYPICAL BIP26 SKETCH

```

_____ #include #include #include
#include #include #include #include // Add these if using onboard sensors: // #include // #include
_____ REFERENCE SKETCHES
ALREADY WRITTEN _____ The
following sketches have been written and tested on this hardware. You can ask Claude to explain,
modify, or extend any of them: Claude Master Prompt Ver A 5 th April 2026. Use to contextualise
your AI Expece, can be applied to other GAI. Page 6 of 6 BIP26_GNSS_DisplayV3.ino
Constellation monitor — colour-coded bar chart showing satellites in view for GPS, Galileo,
GLONASS, BeiDou, QZSS. Also shows date, time, latitude, longitude, altitude.
BIP26_GNSS_QualityGauge.ino Semicircular HDOP speedometer dial — colour coded green to
red. Shows satellites, estimated accuracy, fix type, UTC time. BIP26_PVT_Display.ino Simple
teaching example under 200 lines showing Position, Velocity, Time from TinyGPSPlus. Well
commented to show students how a program is structured. =====
END OF PROMPT =====

```

APPENDIX B NMEA_Reader Program

This tests basic connectivity — see GitHub. Copy this into the Arduino IDE and save with the file name NMEA_Reader.ino (note: .ino is the suffix for Arduino code, and these are the last three letters of Arduino)

```

/* *
=====
* STMP26 SENSATE GPS NMEA Reader - FINAL WORKING LAB REFERENCE
* Target Hardware: TenStar ESP32-S3 + Waveshare LC29H GPS Module
* Recommended Software Environment: Arduino IDE 2.x.x
*
=====
* *  ENGINEERING RULES OF THE ROAD (READ BEFORE FLASHING):
* * 1. THE ENVIRONMENT VARIABILITY FACTOR:
* Depending on whether you are running Windows (and if it's a personal laptop
* or a locked-down institution machine), a Mac, or Linux, the Arduino IDE 2.x
* environment may interact differently with your hardware. If your port
* disappears or throws an "Access Denied" error, check your OS device drivers.
* * 2. THE FIRST LAW OF ENGINEERING SKEPTICISM:
* As an engineer, you must assume absolutely nothing works until YOU have
personally
* tested it and verified the data stream. It does not matter if your lecturer
* swears it works, it does not matter if a generative AI confidently tells you
* it works, and it definitely does not matter if your lab partner says "it
worked
* on my machine five minutes ago." Trust your oscilloscope, trust your serial
* monitor, and verify everything!

```

```
* *
=====
*/

void setup() {
  // Initialize the primary USB serial communication link back to your computer.
  // This allows the ESP32-S3 to send text to the Arduino IDE Serial Monitor.
  Serial.begin(115200);

  // A brief delay to allow the hardware serial interface to stabilize after
  power-up.
  delay(1000);

  // Print diagnostic headers to the Serial Monitor so we know the firmware
  booted successfully.
  Serial.println("=====");
  Serial.println("STMP26 GPS NMEA Reader");
  Serial.println("TenStar ESP32-S3 + WaveShare LC29H GPS");
  Serial.println("=====");
  Serial.println();
  Serial.println("Configuration:");
  Serial.println("  GPS Baud: 115200");
  Serial.println("  RX: GPIO17");
  Serial.println("  TX: GPIO18");
  Serial.println();
  Serial.println("Waiting for GPS data...\n");

  // Initialize Hardware Serial 1 (Serial1) to communicate directly with the
  GPS.
  // CRITICAL PARAMETERS EXPLAINED:
  // - 115200: The default baud rate of the WaveShare LC29H (Note: standard GPS
  modules use 9600).
  // - SERIAL_8N1: 8 data bits, no parity, 1 stop bit (the standard serial
  protocol configuration).
  // - 17: The ESP32 GPIO pin assigned as RX (Receiving data FROM the GPS
  module's TX pin).
  // - 18: The ESP32 GPIO pin assigned as TX (Transmitting data TO the GPS
  module's RX pin).
  Serial1.begin(115200, SERIAL_8N1, 17, 18);

  // Give the serial link to the GPS module a brief moment to open cleanly
  before reading.
  delay(2000);
}
```

```
void loop() {  
  // Check if there are bytes waiting in the hardware buffer received from the  
  GPS module.  
  if (Serial1.available()) {  
  
    // Read the incoming characters from the GPS stream until a newline ('\n')  
    is hit.  
    // This captures exactly one full standard NMEA sentence at a time.  
    String nmea = Serial1.readStringUntil('\n');  
  
    // Strip away any trailing whitespace or hidden carriage return characters  
    (\r)  
    // to keep our print output clean and readable.  
    nmea.trim();  
  
    // Echo the cleaned-up NMEA sentence ($GNGGA, $GNGSV, etc.) over the USB  
    link  
    // directly onto your laptop's Serial Monitor.  
    Serial.println(nmea);  
  }  
}
```